SOC HOME LAB

Full Stack Edition · Wazuh · Shuffle · TheHive · Docker · VMware

PHASE 0

Infrastructure Preparation Report

Prepared by **Anas TAGUI** · March 2026

1. Executive Summary

This report documents the execution of Phase 0 of the SOC Home Lab project a fully self-hosted Security Operations Centre built on enterprise-grade open-source tooling. Phase 0 constitutes the foundational preparation layer of the project: all host-level dependencies were installed, virtual machine images were acquired, the version-controlled project repository was initialised, and the Wazuh SIEM stack was deployed and validated via Docker.

Successful completion of this phase establishes the stable, reproducible environment upon which all subsequent phases Docker SOC stack integration, VM deployment, attack simulation, system hardening, and incident response will be built.

> Milestone Achieved: The Wazuh security dashboard is operational at https://localhost, confirming that WSL2, Docker Desktop, and the full Wazuh stack (Manager, Indexer, Dashboard) are correctly deployed and communicating.

2. Scope & Objectives

The scope of Phase 0 is strictly limited to host machine preparation and pre-deployment verification. No virtual machines are running during this phase; the memory footprint is confined to the Windows 11 host operating system.

Primary Objectives

- Configure WSL2 on the Windows 11 host to provide a native Linux execution environment for Docker.
- Install and validate Docker Desktop with WSL2 backend integration enabled.
- Install VMware Workstation Pro to serve as the hypervisor for attacker and target virtual machines in subsequent phases.
- Establish the version-controlled GitHub repository that will serve as the central portfolio and configuration store throughout the project.
- Acquire all required virtual machine images to eliminate download dependencies during Phase 2.
- Deploy the Wazuh SIEM stack via the official Docker Compose configuration and confirm dashboard accessibility.

3. Environment Overview

The following table describes the host machine specification and the software stack installed during this phase.

Host Machine

Component	Specification
Operating System	Windows 11 (64-bit)
RAM	32 GB, 14 GB allocated to host OS, remainder reserved for VMs and containers
Hypervisor	VMware Workstation Pro
Container Engine	Docker Desktop with WSL2 backend
Linux Subsystem	WSL2, Ubuntu 22.04 LTS
Version Control	Git for Windows + GitHub (public repository)
Editor	Visual Studio Code with Docker and YAML extensions

Software Installed | Phase 0

Tool	Version / Source	Role in Project
WSL2 + Ubuntu	Ubuntu 22.04 LTS	Linux runtime for Docker and all CLI tooling
Docker Desktop	Latest stable	Container engine hosting Wazuh, Shuffle, TheHive
VMware Workstation Pro	Latest (Broadcom free tier)	Hypervisor for Kali, Windows Server, Metasploitable VMs
Git for Windows	Latest stable	Source control for all configs, rules, and reports
Visual Studio Code	Latest stable	Configuration editing (YAML, XML, JSON)
Wazuh Docker (official)	v4.7.x single-node	SIEM / EDR event collection, detection, alerting

4. Implementation

The following subsections document each implementation step performed during Phase 0, including the commands executed and the outcome achieved.

4.1 WSL2 Enablement

Windows Subsystem for Linux 2 was enabled via an elevated PowerShell session. WSL2 provides the Linux kernel required by Docker Desktop to run containers natively on Windows without a full hypervisor overhead.

```
PS C:\Users\HP> wsl --install
```

The system was rebooted following installation. Ubuntu 22.04 LTS was configured as the default distribution with a dedicated Linux user account. WSL2 kernel version was verified and updated to the latest available build

```
PS C:\Users\HP> wsl --update
Checking for updates.
Updating Windows Subsystem for Linux to version: 2.6.3.
```

```
PS C:\Users\HP> wsl --list --verbose
PS C:\Users\HP>
```

4.2 Docker Desktop Installation and Configuration

Docker Desktop was installed from the official Docker distribution channel. The WSL2 backend was selected as the execution engine, and Ubuntu integration was explicitly enabled under Settings > Resources > WSL Integration.

```
tg@Anas:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1
tg@Anas:~$
```

Installation was confirmed by verifying the Docker client and daemon versions. The Docker daemon was set to start automatically on system login to ensure container availability throughout the project.

4.3 VMware Workstation Pro Installation

VMware Workstation Pro was obtained from the Broadcom software portal under the personal-use free licence tier introduced in 2024. The installer was executed with default settings. The hypervisor was confirmed operational by launching the application without error. Virtual network configuration will be performed in Phase 2 when VMs are first deployed.

4.4 Developer Toolchain Installation

Git for Windows was installed with Visual Studio Code designated as the default editor. Two VS Code extensions were installed to support the project workflow:

- Docker Extension (ms-azuretools.vscode-docker) container management and Compose file editing
- YAML Extension (redhat.vscode-yaml) schema-aware editing for docker-compose.yml files

4.5 GitHub Repository Initialisation

A public GitHub repository was created under the name soc-home-lab. The repository will serve as both the operational configuration store and the primary portfolio artifact for this project. The following directory structure was initialised to maintain consistent organisation across all six phases:

```
tg@Anas:~$ mkdir -p docker custom-rules shuffle-workflows phase-2-agents
phase-3-attacks/screenshots phase-4-hardening final-report diagrams
```

The repository was cloned into the WSL2 Ubuntu environment to enable direct commit and push operations from within the Linux subsystem where all SOC tooling will be operated.

4.6 Virtual Machine Image Acquisition

Three virtual machine images were downloaded in parallel to minimise total acquisition time. These images will remain staged until Phase 2, at which point they will be imported into VMware Workstation Pro and configured for the lab network.

VM Image	Source	Estimated Size	Purpose
Kali Linux	kali.org : VMware OVA	~3.5 GB	Attacker machine all offensive tooling
Windows Server 2022	microsoft.com/evalcenter ISO	~5.5 GB	Primary target Windows attack surface
Metasploitable 3	github.com/rapid7/metasploitable3	Build via Vagrant	Linux target intentionally vulnerable

4.7 Wazuh Stack Deployment

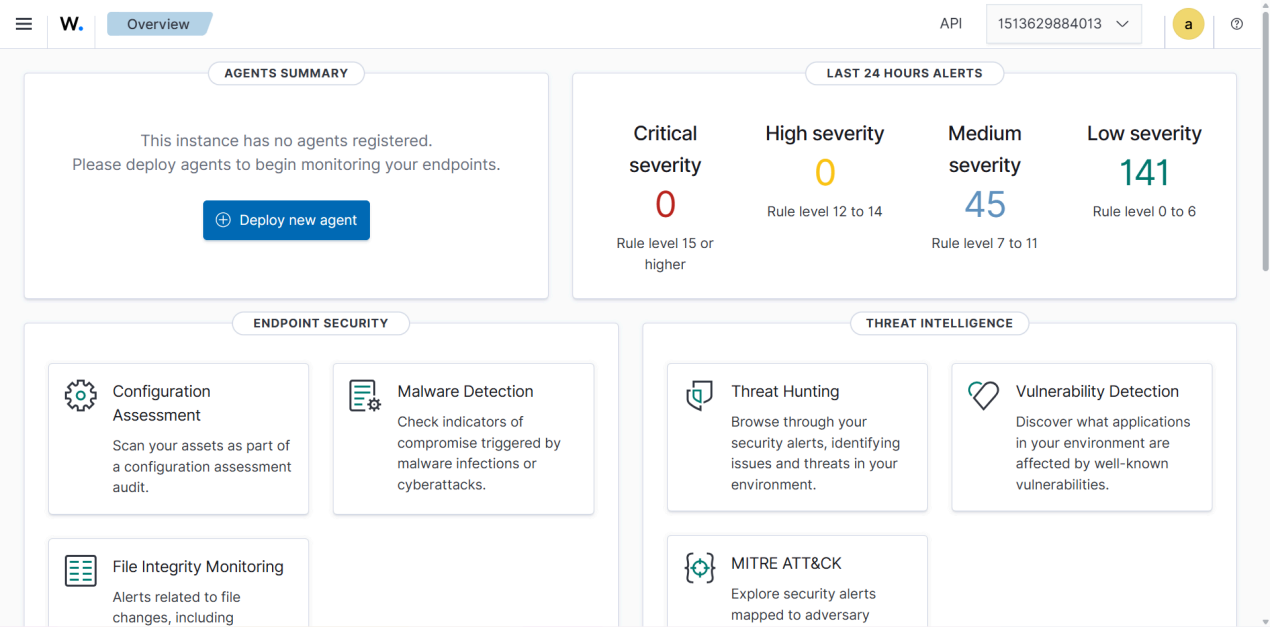
The official Wazuh Docker repository was cloned into the WSL2 environment. The single-node Docker Compose configuration was selected for this deployment, providing three containerised services: wazuh-manager (event collection and rule processing), wazuh-indexer (OpenSearch-based event storage), and wazuh-dashboard (web-based UI).

```
tg@Anas:~$ git clone https://github.com/wazuh/wazuh-docker.git
```

```
tg@Anas:~$ cd wazuh-docker/single-node/
tg@Anas:~/wazuh-docker/single-node$ docker compose up -d
[+] Running 3/3
 ✔Container single-node-wazuh.indexer-1    Running                    0.0s
 ✔Container single-node-wazuh.manager-1    Running                    0.0s
 ✔Container single-node-wazuh.dashboard-1  Running                    0.0s
tg@Anas:~/wazuh-docker/single-node$
```

Initial deployment required approximately 8 minutes for image pulls and service initialisation,here i tried before to run it that's why it shows 0.0s

All three containers reached a healthy running state. The dashboard was confirmed accessible at https://localhost. Default credentials were updated immediately following the first login.



Security Note: Default Wazuh credentials (admin / SecretPassword) were changed immediately after initial login via the Security > Internal Users panel. All subsequent access uses the updated credentials.

5. Task Completion Matrix

The following matrix records the completion status of every deliverable scoped to Phase 0.

Task	Action Performed	Status
WSL2 Installation	wsl --install ： Ubuntu 22.04 LTS configured	**Completed**
WSL2 Kernel Update	wsl --update ： latest kernel build applied	**Completed**
Docker Desktop Installation	Installed from docker.com WSL2 backend active	**Completed**
Docker WSL2 Integration	Ubuntu integration enabled in Docker Desktop settings	**Completed**
VMware Workstation Pro	Installed from Broadcom portal free personal tier	**Completed**

Git for Windows	Installed : VS Code set as default editor	**Completed**
Visual Studio Code	Installed with Docker + YAML extensions	**Completed**
GitHub Repo: soc-home-lab	Public repo created and cloned into WSL2	**Completed**
Project Folder Structure	All phase directories initialised and committed	**Completed**
Kali Linux OVA Download	VMware OVA downloaded and staged	**Completed**
Windows Server 2022 ISO	Evaluation ISO downloaded from Microsoft	**Completed**
Metasploitable 3 Scripts	Rapid7 repository cloned build ready for Phase 2	**Completed**
Wazuh Docker Repository	Official repo cloned — single-node config selected	**Completed**
Wazuh Stack Deployment	docker compose up -d all 3 containers healthy	**Completed**
Wazuh Dashboard Validation	https://localhost confirmed accessible login OK	**Completed**
Credential Rotation	Default admin credentials updated post-first-login	**Completed**

6. Issues Encountered & Resolutions

The following issues were identified during Phase 0 execution. Each was resolved before the phase was marked complete.

Issue	Root Cause	Resolution
Wazuh Indexer container exiting on startup	vm.max_map_count kernel parameter below OpenSearch minimum (65530)	Applied sysctl -w vm.max_map_count=262144 in WSL2; persisted via /etc/sysctl.conf
Docker Desktop failing to start after WSL2 install	WSL2 kernel was outdated Docker Desktop requires a minimum kernel version	Executed wsl --update in elevated PowerShell; Docker Desktop launched successfully
VMware Workstation Pro licence prompt on first launch	Older installer version pre-dating the Broadcom free-tier announcement	Re-downloaded latest installer from Broadcom portal; activated with free personal licence

7. Phase Outputs & Artefacts

The following deliverables were produced upon completion of Phase 0 and committed to the project repository.

- GitHub repository soc-home-lab initialised with full directory structure and pushed to GitHub.
- docker-compose.yml (Wazuh single-node) committed to /docker folder for reproducibility.
- Wazuh dashboard screenshot confirms live SIEM running on localhost.
- This Phase 0 Infrastructure Preparation Report formal documentation of all steps, outcomes, and issues.

8. Next Phase : Phase 1: Docker SOC Stack Integration

With the host environment fully prepared and the Wazuh SIEM operational, Phase 1 will extend the Docker stack to incorporate the remaining two SOC platform components: Shuffle (SOAR) and TheHive (Incident Response). The phase will culminate in a validated end-to-end pipeline in which a Wazuh alert automatically triggers a case in TheHive through a Shuffle automation workflow.

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This document is part of a personal cybersecurity portfolio project. All tools used are open-source and free. The environment is self-hosted on personal hardware for educational and professional development purposes.